BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 225 · OOP: testing and debugging

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Insert 12 into a BST node 10 with empty right child. Where does it go?

Q2. What is a reference?

Q3. What is a local variable?

Q4. If an object's Balance=20 and Deposit(5) correctly adds five, what is Balance?

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I Do · Read new knowledge

Key vocabulary

encapsulation — Hide implementation details and control state through methods/properties.

constructor — Initialises a new object.

field / property — Stores or exposes object state.

breakpoint — Pauses execution at a selected code line.

boundary test — Checks values at or just beyond a permitted limit.

New learning

Inspect class state — The object has a Balance property initially set by its constructor.

Trace method inputs — Deposit(5) passes amount=5. Since amount>0, Balance+=5 executes.

Debug the outcome — Expected Balance=25. If unchanged, check whether the code assigns to Balance or merely calculates a temporary value.

Worked example

Find and correct the fault when AddPoints(3) fails to update Score=7.

1. Predict first — The local newScore becomes 10, but the field Score remains 7.

2. Model the reasoning — Replace the local-only calculation with Score += points;.

3. Model the reasoning — Retest: after AddPoints(3), Score should equal 10.

We Do · Practise with guidance

A Counter starts at 2. Increment() contains `Count + 1;`. Explain the problem and fix it.

```
public void Increment() {
  Count + 1; // invalid statement in C#
}
```

Predict first

Count + 1 does not assign a new value and is not a valid standalone statement in C#.

Model the reasoning

Correct it to Count++; or Count += 1;. With Count=2, the resulting Count is 3.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

An object starts at Count=4 and runs Increment() which performs Count += 1. What is Count after three calls?

Question 2

Correct `public void SetAge(int age) { int Age = age; }` where Age is an existing property. Explain why it fails.

Question 3

A BankAccount.Withdraw(amount) rejects only amounts <=0, but permits overdrafts. Add validation and give boundary tests for Balance=50.

Exit Ticket · Answer before checking

A method may calculate a correct new value without actually updating the object's property.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. As the right child of 10.

Q2. A value that identifies an object rather than copying the object itself.

Q3. A variable declared inside a method or block with limited scope.

Q4. 25.

You Do pages · Independent questions

Q1. [4] 5 after first [1], 6 after second [1], 7 after third [1]; each call updates the same object's field [1].

Q2. [6] `int Age` declares a new local variable [2], shadowing rather than updating the member [1]; use `this.Age = age;` [2]; test SetAge(16) sets property to 16 [1].

Q3. [6] Reject amount<=0 [1] and amount>Balance [1]; valid update Balance-=amount [1]. Boundary tests: Withdraw(50) gives 0 [1], Withdraw(51) is rejected [1], Withdraw(0) is rejected [1].

Exit Ticket · Model answer

True. A missing assignment can leave object state unchanged.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick and rewrite mistakes.